

DELAWARE STATE UNIVERSITY
Department of Education
Fall 2026

I. COURSE DETAILS

Course Number: EDUC 331A
Course Name: Methods of Teaching Science in Elementary and Middle Level
CRN: 16179
Credit: 3
Delivery Modality: Hybrid (In-person, Synchronous, and Asynchronous)

Instructor Name: SaeYeol Yoon, Ph.D.
Instructor Email: syoon@desu.edu
Instructor Phone: 6726
Instructor Office Location: EH245
Office Hours: Tuesday/Wednesday 11 am (Monday/Thursday Virtual) or by appointment

Class Location: EH201A
Meeting Day(s): Tuesday & Thursday
Meeting Time(s): 03:00 – 04:15 pm
Preferred Communication Method: email
Expected Faculty Response Time: 48 hours

II. Required Texts, Materials, and Technology:

Required Textbooks:

- Windschitl, M., Thompson, J., & Braaten, M. (2018). *Ambitious science teaching*. Harvard Education Press.

Recommended Resources: None required. (Optional readings or supplementary articles will be provided digitally via Canvas.)

Technology Requirements: Reliable internet access, & A computer, laptop, or tablet capable of browsing the web and viewing digital course materials

Required Software/Applications: NA

Educational Platforms Used: Canvas

III. COURSE DESCRIPTION

The course includes basic teaching strategies, materials, and evaluative techniques for elementary school teachers. Classroom observation and teaching, unit and lesson development are required. Attention will also be given to developing science programs made from materials at hand using a standard text. Early field experience is required (10 hours). Prerequisites: PSED 201. Admission to the Teacher Education Program is required.

IV. COURSE RATIONALE

This course is designed to provide you with meaningful and practical learning experiences that will prepare you to create effective elementary and middle level science learning environments for students and to construct a vision of yourself as a teacher of science. Several driving questions guide this methods course:

- How do elementary and middle level students learn science?
- What classroom environments facilitate students' science learning?
- What should teachers know and be able to do to design and foster effective science learning environments?

V. PROFESSIONAL STANDARDS

All course assignments and activities in the Department of Education align with CAEP Standard 1: Content and Pedagogical Knowledge – The provider ensures that candidates develop a deep understanding of the critical concepts and principles of their discipline and, by completion, are able to use discipline-specific practices flexibly to advance the learning of all students toward attainment of college- and career-readiness standards

RECAP Demonstrations

1: Learning Environment *Creating inclusive, safe, and supportive classrooms*

Teacher candidates demonstrate the ability to:

- Build positive relationships with students grounded in trust, empathy, and high expectations
(ED-SLO 2; InTASC 3)
- Affirm student identity, culture, and lived experience through culturally responsive practice
(ED-SLO 2, ED-SLO 7; InTASC 2)
- Promote student voice, agency, and belonging
(ED-SLO 2; InTASC 3)
- Implement proactive and restorative classroom management strategies
(ED-SLO 2; InTASC 3)
- Reflect on personal values, bias, and positionality to support equitable learning environments
(ED-SLO 7; InTASC 2, 9)

2: Engagement in Learning *Designing and implementing instruction that actively involves all learners*

Teacher candidates demonstrate the ability to:

- Design developmentally appropriate, standards-aligned learning experiences
(ED-SLO 1; InTASC 1 – Learner Development, InTASC 7 – Planning)
- Use discipline-specific strategies and academic language effectively
(ED-SLO 3; InTASC 4, InTASC 5)
- Use student-centered and inquiry-based instructional strategies
(ED-SLO 1, ED-SLO 4; InTASC 8 – Instructional Strategies)
- Differentiate instruction to meet diverse learner needs
(ED-SLO 1; InTASC 2 – Learning Differences)
- Engage learners through collaboration, questioning, and academic discourse
(ED-SLO 4; InTASC 8)
- Connect instruction to real-world contexts and student experiences to support application, transfer, and deeper understanding of content
(ED-SLO 3, ED-SLO 4; InTASC 5 – Application of Content)
- Integrate instructional technology intentionally and ethically
(ED-SLO 6; InTASC 8)
- Use digital tools to support access, engagement, and differentiation
(ED-SLO 6; InTASC 8)
- Adjust instruction responsively based on student engagement and understanding
(ED-SLO 1; InTASC 6 – Assessment, InTASC 8)

3: Maximizing Learning *Ensuring deep understanding, growth, and academic progress*

Teacher candidates demonstrate the ability to:

- Use formal and informal assessments to monitor student learning
(ED-SLO 5; InTASC 6 – Assessment)
- Analyze student data to inform instructional decisions
(ED-SLO 5; InTASC 6)
- Provide timely, specific, and actionable feedback
(ED-SLO 5; InTASC 6)
- Support productive struggle with appropriate scaffolds
(ED-SLO 5; InTASC 1, InTASC 8)
- Adjust instruction based on evidence of student progress
(ED-SLO 5; InTASC 6, InTASC 7 – Planning)
- Promote metacognition and student reflection to support learning growth
(ED-SLO 5; InTASC 6)

4: Professional Responsibilities *Professional practice, collaboration, and advocacy*

Teacher candidates demonstrate the ability to:

- Reflect critically on teaching practice using evidence and feedback
(ED-SLO 8; InTASC 9 – Professional Learning & Ethical Practice)
- Engage in continuous professional learning and improvement
(ED-SLO 8; InTASC 9)
- Demonstrate professionalism, reliability, and accountability
(ED-SLO 8; InTASC 9)
- Collaborate effectively with mentors, colleagues, and support staff
(ED-SLO 8; InTASC 10 – Leadership & Collaboration)
- Advocate ethically and equitably for learners
(ED-SLO 9; InTASC 9, InTASC 10)
- Engage families as partners in student learning
(ED-SLO 9; InTASC 10)
- Build culturally responsive family-school relationships
(ED-SLO 9; InTASC 10)
- Collaborate with community stakeholders to support learner growth
(ED-SLO 9; InTASC 10)

Gholdy Muhammad's Five Pursuits Framework

The Department of Education is committed to instructional practices that support the holistic development of learners through the integration of Gholdy Muhammad's Historically Responsive Literacy Framework. Course content, assignments, and learning experiences may incorporate the following five pursuits:

- **Identity Development**
Supports learners in developing a positive understanding of themselves, their cultures, communities, histories, and lived experiences while affirming the identities of others.
- **Skill Development**
Focuses on building academic, disciplinary, technological, and professional skills necessary for success in educational and professional settings.
- **Intellectual Development**
Encourages learners to think critically, analyze information, solve problems, and deepen understanding through inquiry, reflection, and application of knowledge.
- **Criticality**
Promotes awareness of power, equity, bias, and social issues while encouraging learners to question systems, advocate for justice, and engage in meaningful conversations about the world around them.
- **Joy**
Cultivates engagement, creativity, curiosity, wellness, and meaningful learning experiences that honor student voice, promote belonging, and foster a positive learning environment.

ISTE STANDARDS FOR EDUCATORS**Empowered Professional**

Standard 1: Learning – Educators continually improve their practice by learning from and with others and exploring proven and promising practices that leverage technology to improve student learning.

Standard 2: Leader – Educators seek out opportunities for leadership to support student empowerment and success and to improve teaching and learning.

Standard 3: Citizen – Educators inspire students to positively contribute to and responsibly participate in the digital world.

Learning Catalyst

Standard 4: Collaborator – Educators dedicate time to collaborate with both colleagues and students to improve practice, discover and share resources and ideas, and solve problems.

Standard 5: Designer – Educators design authentic, learner-driven activities and environments that recognize and accommodate learner variability.

Standard 6: Facilitator – Educators facilitate learning with technology to support student achievement of the 2016 ISTE Standards for Students.

Standard 7: Analyst – Educators understand and use data to drive their instruction and support students in achieving their learning goals.

Education Department SLOs

ED-SLO1 – Developmentally Appropriate and Equitable Instruction

Teacher candidate will demonstrate an understanding of how to design and implement developmentally appropriate, equitable, challenging, and differentiated learning experiences.

ED-SLO2 – Safe and Inclusive Learning Environments

Teacher candidate will establish a safe, inclusive, and respectful learning environment for all students to learn.

ED-SLO3 – Content Knowledge and Student Identity Connection

Teacher candidate will demonstrate pedagogical competence, as well as the understanding of intellectual history of the content they teach and how it connects to student identities.

ED-SLO4 – Critical Thinking and Problem-Solving

Teacher candidate will develop learning experiences that make the content relevant by engaging learners in criticality, critical thinking, creativity, and problem solving.

ED-SLO5 – Data-Driven Assessment

Teacher candidate will apply formal and informal assessment strategies to evaluate and assess individual learning needs and to ensure the continuous intellectual and social development of the learner.

ED-SLO6 – Curriculum, Instruction, and Technology Integration

Teacher candidate will organize curriculum, instruction, and assessment while leveraging technology to help diverse learners meet rigorous learning goals.

ED-SLO7 – Bias Awareness and Inclusive Strategies

Teacher candidate will understand awareness of their own personal values and biases to select appropriate instructional strategies.

ED-SLO8 – Professional Growth and Self-Evaluation

Teacher candidate will demonstrate a commitment to life-long learning by engaging in professional learning and collaborative practices and uses evidence to continually evaluate their practice to improve teaching and learning.

ED-SLO9 – Community and Family Advocacy

Teacher candidate will advocate using authentic engagement practices, for learners, families, and the community in an ethical and equitable way to ensure learner growth.

Delaware State University SLOs

DSU-SLO-1 – competent communicators

DSU-SLO-2 – effective inquirers, critical thinkers, and problem-solvers able to use appropriate quantitative and qualitative information

DSU-SLO-3 – ethical, collaborative, and productive citizens of a complex, diverse world

DSU-SLO-4 – independent learners able to integrate knowledge and technology to achieve personal and professional success

Graduate programs also include a fifth learning goal to prepare our graduates for:

DSU-SLO-5 – outreach and service

[CLICK HERE](#). Add additional standards related to the course

VI. KEY COURSE OBJECTIVES/OUTCOMES

Course Objective	RECAP Demonstrations	ISTE	DSU-SLOs	*Program SLOs	Program Standards	5 Pursuits
Present and defend your beliefs about elementary and middle level science teaching and learning	4	1, 5	1, 2	ELEM, MLE	NSTA 2, 5	Identity, Criticality
Use appropriate resources and strategies to develop an in-depth understanding of the science concepts you will teach	2	1, 2, 5, 6	2, 4	ELEM, MLE	NSTA 1, 2, 5	Skill, Intellectual

Draw upon theories of learning to plan appropriate science learning experiences	2	1, 5	2, 4	ELEM, MLE	NSTA 2, 3, 5	Intellectual
Elicit and utilize students' ideas about science concepts to plan learning experiences	2	1, 2, 6	2, 3	ELEM, MLE	NSTA 1, 2	Identity, Intellectual
Recognize the characteristics of teaching science as inquiry	2	5, 6	2, 4	ELEM, MLE	NSTA 2, 5	Intellectual, Joy
Promote classroom discourse that reflects the nature of science and supports students' science learning	2	4, 5, 6	1, 3	ELEM, MLE	NSTA 2, 3, 5	Skill, Criticality
Systematically collect assessment data about students' science learning and use it to analyze the effectiveness of your teaching and prepare for future instruction	3	5, 7	2, 4	ELEM, MLE	NSTA 2, 3, 4	Skill, Intellectual
Connect science to other areas of the curriculum and use technology to enhance students' science learning	2	2, 4, 5	4	ELEM, MLE	NSTA 2, 3, 4, 5	Skill
Determine ways of promoting "science for all" in your classroom	1	1, 2, 3	3	ELEM, MLE	NSTA 3, 5	Criticality, Identity
Plan and teach lessons based upon appropriate instructional models that you learn in this course	2	5, 6	2, 4	ELEM, MLE	NSTA 2, 3, 4, 5	Skill, Joy
Critically reflect on your own and others' teaching, noting areas of strength as well as needed improvement	4	4, 5, 6	2	ELEM, MLE	NSTA 2, 5	Intellectual, Criticality

*Program SLOs Legend: Elementary Education=ELEM; Middle Level Education=MLE

VII. TOPICAL OUTLINE & TIMELINE SCHEDULE

Week	Module	Topic/Reading/Focus	Assignments and Learning Activities
Week 1	Module 1	Naive theories of learning, teaching, and science; launching an open investigation	Bread Mold Growth investigation; co-teaching pairs assigned; Personal Definition

Week	Module	Topic/Reading/Focus	Assignments and Learning Activities
			of Learning; Teaching Science 3-Minute Brainstorm
Week 2	Module 1	Peer teaching without a framework ("the before"); perspective-taking in learning	Baseline Co-Teaching (video recorded); Fish is Fish; Exploring What Makes a Good Science Lesson; Concept Map + Video Reflection
Week 3	Module 2	Levels of inquiry; Nature of Science	Cool-It Labs jigsaw; Nature of Science reflection
Week 4	Module 2	NGSS three-dimensional learning; lesson-design frameworks (5E Model)	NGSS structure worksheet; 5E Model Lesson Analysis
Week 5	Module 3	Eliciting student thinking	RSST data collection and debrief
Week 6	Module 3	Productive talk moves in practice	Talk Moves microteach (video recorded)
Week 7	Module 4	Field-readiness logistics (no class — university midterm week)	Residency toolkit/logistics finalization; materials & safety plan
Week 8	Module 4	Using models formatively; equity in curriculum	Draw What You Think; Whose Voices Are in This Unit?; peer feedback
Week 9	Module 5	Field Residency I — Supporting ongoing changes in thinking	Enact lessons in placement; practice log
Week 10	Module 5	Field Residency II — Analyzing real teaching	Peer Teaching Lesson Plan Analysis; Seeing Science Everywhere
Week 11	Module 6	Argumentation and evidence-based reasoning	K-2 Argumentation Series; Reflective Teaching Dialogues
Week 12	Module 7	Planning from a performance expectation; designing for practice	3D Lesson Plan Tasks 1–4; first draft
Week 13	Module 7	Peer review; capstone peer teaching	Peer feedback round; Capstone co-teaching (video recorded)
Week 14	Module 8	Comparing revised practice to baseline (async — Thanksgiving week)	Teach It Again — co-teaching pairs re-teach baseline (video recorded)
Week 15	Module 8	Lesson study; course synthesis	Fresh Eyes lesson study; final side-by-side comparison and sharing

VIII. INSTRUCTIONAL METHODS/BEST PRACTICES

A combination of lecture, discussion, and cooperative learning techniques is supplemented by interactive, computer-based instruction, including online learning modules, peer teaching, and structured field observation in a K–8 classroom. Students should be comfortable using the internet and the course learning management system, as assignments and learning modules will be distributed and submitted accordingly.

IX. EVALUATION/ASSESSMENT

Grading Policy

Course grades are based on the evaluation methods listed in the Grades Breakdown below, which measure candidates' developing content knowledge, pedagogical skill, and professional dispositions related to teaching elementary and middle level science. Assignments and assessments are weighted according to their depth and scope, and feedback is provided in a

timely manner to support candidates' ongoing growth. Students must earn a C or higher to successfully complete this course, consistent with Teacher Education Program requirements.

**Students must receive a C or higher to successfully complete this course.*

Grades Breakdown

Evaluation Method (i.e.: Tests, Assignments, Projects, etc.)	Percentage/Points
Attendance & Participation	10%
In-Class Activities & Online Learning Modules	50%
Midterm Essay	15%
Peer Teaching & Reflection	25%

Grading Scale

A = 90–100% B = 80–89% C = 70–79% D = 60–69% F = 59% & below

X. COURSE POLICIES & PROCEDURES

Students should complete course evaluations sent by the University each semester.

Attendance

Regular attendance and active participation are required in all course formats. For face-to-face and hybrid sessions, students are expected to arrive on time, remain for the full session, and participate in discussions, group work, and hands-on science activities — much of this course's learning happens through collaborative, in-person practice that cannot be replicated asynchronously. More than two unexcused absences from face-to-face or hybrid sessions may lower the final grade by one letter grade for each additional absence; three or more may result in course failure, at the instructor's discretion. For online modules, "attendance" is defined as timely, substantive engagement with posted materials, discussion boards, and assignments by the stated deadlines — logging in without contributing does not count as attendance. Excused absences (illness, family emergency, religious observance, university-sanctioned activity) require advance notice when possible and documentation upon return; students are responsible for arranging to make up missed content and coordinating with the instructor within one week of the absence. Professionalism is part of attendance: students are expected to communicate proactively about anticipated absences, arrive prepared, and treat class time as they would a professional obligation in a school setting.

Make-Up/Late Assignments

Assignments are due at the date and time specified in the course schedule. Late work is accepted only in cases of documented excused absence (illness, emergency, religious observance, university-sanctioned event) and must be submitted within one week of the original due date unless other arrangements are made in advance with the instructor. Undocumented late submissions will be penalized 10% per day late, up to a maximum of five days, after which the assignment will receive a zero. Students anticipating a conflict with a due date should contact the instructor before the deadline, not after. Make-up work for missed in-class activities (labs, group science investigations, peer teaching demonstrations) will be assigned at the instructor's discretion and may take a different form than the original activity, since some collaborative or hands-on experiences cannot be fully replicated individually. Extensions for extenuating circumstances (family emergencies, documented health issues, etc.) will be considered on a case-by-case basis; students should reach out as soon as they are aware of the need.

Professionalism

As future educators, students are expected to model the professional conduct they will be expected to demonstrate in a K-12 classroom. For online and hybrid components, students must keep cameras on during synchronous sessions unless a documented accommodation or specific circumstance has been discussed in advance with the instructor — this course involves discussion-based and collaborative work that depends on visible engagement. Microphones should be unmuted for discussion and muted otherwise to minimize background noise; students should participate from a quiet, distraction-free setting appropriate for a professional class meeting. Written communication (discussion posts, emails, group chats) should be professional in tone and free of the kind of shorthand appropriate for texting. Clinical/field placement attendance is mandatory and governed by the same standards as employment: students must attend all scheduled placement days, arrive on time, and notify both the course instructor and the cooperating teacher/school directly (not just the university) in the event of any absence, ideally before the start of the school day. Absences from clinical placement must be made up in coordination with the cooperating teacher and program coordinator, as missed clinical hours can jeopardize certification requirements. Repeated unprofessional conduct — in class, online, or in the field placement — may be referred to the program's professional dispositions review process.

Communication and Digital Professionalism

Professional communication and responsible digital engagement are expected in this course. Students are expected to communicate respectfully and professionally with classmates, instructors, school partners, and guest speakers in all face-to-face and virtual interactions. This includes email communication, discussion boards, video conferencing platforms, collaborative documents, and other digital learning environments.

Students should:

- Use respectful and appropriate language in all communications.
- Maintain professional behavior during class discussions and virtual meetings.
- Regularly monitor university email and Blackboard announcements for course updates.
- Use constructive and inclusive communication that supports a positive learning environment.
- Respect the privacy, perspectives, and contributions of others.
- Demonstrate responsible and ethical use of technology and digital resources.
- Follow FERPA and confidentiality expectations when discussing students, schools, field experiences, or classroom observations.

For virtual or hybrid learning environments, students are expected to participate appropriately, minimize distractions, and engage professionally during synchronous sessions. Inappropriate, disruptive, or disrespectful behavior in digital or in-person learning spaces may impact participation and professionalism evaluations.

As future educators and educational professionals, students are encouraged to develop digital professionalism skills that reflect ethical, inclusive, and responsible engagement in academic and professional settings.

Universal Design for Learning (UDL) Statement

This course incorporates Universal Design for Learning (UDL) principles to create an inclusive and accessible learning environment. Course materials, activities, and assessments are designed to provide multiple ways for students to engage with content, participate in learning, and demonstrate understanding. Students are encouraged to communicate with the instructor regarding any learning supports or accommodations that may contribute to their success.

XI. UNIVERSITY POLICIES & PROCEDURES

GENERATIVE ARTIFICIAL INTELLIGENCE (AI) POLICY

Use of AI tools is permitted throughout the process of working on assignments in this course, provided that it is acknowledged and cited appropriately. Students may use such tools to brainstorm, outline, edit, expand, or otherwise modify student-generated work. Students are responsible for assessing the accuracy and value of AI outputs used in their work, as well as for the quality of final work products. Note that other courses may institute other policies governing the use and misuse of AI tools and that it is the student's responsibility to follow the rules for each course.

AMERICANS WITH DISABILITIES ACT (ADA) AND SECTION 504 OF THE REHABILITATION ACT OF 1973

The University is committed to full compliance with the Americans With Disabilities Act of 1990 (ADA) and Section 504 of the Rehabilitation Act of 1973, which prohibit discrimination against qualified persons with disabilities, as well as other federal and state laws pertaining to individuals with disabilities. Under the ADA and its amendments, a person has a disability if he or she has a physical or mental impairment that substantially limits a major life activity. The ADA also protects individuals who have a record of a substantially limiting impairment or who are regarded as disabled by the institution whether qualified or not. A substantial impairment is one that significantly limits or restricts a major life activity such as hearing, seeing, speaking, breathing, performing manual tasks, walking or caring for oneself.

Students requesting accommodations due to a disability at Delaware State University need to present a current Accommodation Letter to faculty before accommodations can be made. While there is no deadline to submit an Accommodation Letter, accommodations will not be put in place retroactively. Accommodation Letters are issued by the Center for Disability Resources (CDR). For additional information, visit the [CDR website](#) or contact CDR directly: (email) cdr@desu.edu, or (office) Education and Humanities Building, Room 123.

ACADEMIC DISHONESTY

Academic dishonesty will not be tolerated. Unless otherwise indicated by the instructor in connection with group work, all work must be individual. Evidence of collusion (working with another student) or plagiarism (use of another's ideas, data, and statements without acknowledgment and/or extensive use of another's ideas, data and statements with only minimal acknowledgment) will lead to the procedures set up by the Delaware State University for academic dishonesty. A student may be brought before the academic review committee, receive an F for the course, or be expelled from Delaware State University. Any student who is caught cheating on an exam, quiz, test or assignment will receive zero points and an F for the course.

OTHER POLICIES

Students are encouraged to visit the University website to view additional policies and procedures.

<https://www.desu.edu/about/administration/university-policies-procedures>

XII. DIVERSITY, EQUITY, INCLUSION, AND BELONGING

Students in this course are expected to be aware of the ISTE (International Society for Technology in Education) standards and infuse instructional practices and technology into all major assignments and into their instructional units. Their presentations are to be prepared using multi-media. Students will interact with teachers and students of diverse backgrounds in inclusive settings. The publications utilize a wide range of professional IT concepts that follow the latest international trends.

XIII. CAMPUS RESOURCES

- [Writing Resources for Students](#)
- [Counseling Services](#)
- [Library Services](#)
- [Tutoring Services](#)
- [Technology Support](#)
[Canvas Login](#)
[Watermark Support](#)

Students must submit Watermark assignments by clicking the link in the associated Canvas course. If there is not a link available, students should contact the course instructor.

XIV. RESOURCES, RESEARCH, AND REFERENCES

Bybee, R. W., Carlson-Powell, J., & Trowbridge, L. W. (2008). Teaching Secondary School Science: Strategies for Developing Scientific Literacy, Prentice Hall.

Hand, B., Norton-Meier, L., Staker, J., & Bintz, J. (2009), Negotiating Science: The Critical Role of Argument in Student Inquiry, Portsmouth, NH: Heinemann.

Michales, S., Shouse, A. W., & Schweingruber, H. A. (2008). Ready, Set, Science! Washington, DC: The National Academies Press.

National Research Council (2011). Successful K-12 STEM Education: Identifying Effective Approaches in Science, Technology, Engineering, and Mathematics. Washington, DC: The National Academies Press.

National Research Council. (2012). A Framework for K-12 Science Education: Practices, Crosscutting Concepts, and Core Ideas. Retrieved from <http://www.nap.edu/>

National Research Council (2013). Next Generation Science Standards, Washington, DC: The National Academies Press. Retrieved by <http://www.nextgenscience.org/>

OTHER HELPFUL RESOURCES AND PUBLICATIONS

American Association for the Advancement of Science (AAAS): <http://www.aaas.org/>

APA Formatting and Style Guide - The OWL at Purdue:
<http://owl.english.purdue.edu/owl/resource/560/01>

APA Official Site: How to Cite Information From the Internet and the World Wide Web:
<http://www.apastyle.org/electref.html>

Centers for Ocean Sciences Education Excellence (COSEE-West):
<http://www.usc.edu/org/cosee-west/index.html>

Council for Exceptional Children (CEC):
<http://www.cec.sped.org//AM/Template.cfm?Section=Home>

Creating Conditions for Learning: COMP - Classroom Organization Management Program:
<http://www.comp.org/>

Delaware Science Standards, Retrieved from:
http://www.doe.k12.de.us/infosuites/staff/ci/content_areas/files/science/dapa_ScienceStandards.pdf

Education Resources Information Center (ERIC): <http://www.eric.ed.gov/>

Gateway to Educational Materials (GEM): <http://www.thegateway.org/>

Inquiry and the National Education Standards A Guide for Teaching and Learning:
http://books.nap.edu/html/inquiry_addendum/preface.html

Laboratory Safety: <http://www.labsafety.org/>

Materials Safety Data Sheet (MSDS) Search: <http://www.msdsearch.com/>

National Association of Special Education Teachers: <http://www.naset.org/>

National Board for Professional Teaching Standards (NBPTS): <http://www.nbpts.org/>

National Board for Professional Teaching Standards (NBPTS) Five Core Propositions:
http://www.nbpts.org/the_standards/the_five_core_propositions

National Science Education Leadership Association (NSELA): <http://www.nsela.org/index.html>

National Science Teachers Association (NSTA): <http://www.nsta.org/>

Science Program Improvement Review: <http://www.nsta.org/main/pdfs/2004AnnualReport.pdf>

Standards for Science Teacher Preparation:

<http://www.nsta.org/main/pdfs/NSTASTandards2003.pdf>

Teacher Performance Assessment -CalTPA Model Tasks: <http://www.ctc.ca.gov/educator-prep/TPA-tasks.html>

Teaching Performance Expectations (TPEs): <http://www.ctc.ca.gov/educator-prep/TPA.html>

The Association for Supervision and Curriculum Development (ACSD):

<http://www.ascd.org/portal/site/ascd>

The Educator's Reference Desk-Science Lesson Plans: <http://www.eduref.org/cgi-bin/lessons.cgi/Science>

The National Academies Advisers to the Nation on Science, Engineering and Medicine:

<http://www.nationalacademies.org/about/>

The Science, Technology, Engineering and Mathematics (STEM) Education Institute:

<http://k12s.phast.umass.edu/stem/>

Underwriters Laboratory Inc. (UL): <http://www.ul.com/>

U.S. Department of Labor Occupational Safety & Health Administration (OSHA):

<http://www.osha.gov/>

U.S. Department of Education: <http://www.ed.gov/>

U.S. Environmental Protection Agency (EPA): <http://www.epa.gov/>

WestEd: <http://www.wested.org/cs/we/print/docs/we/home.htm>geosociety.org/